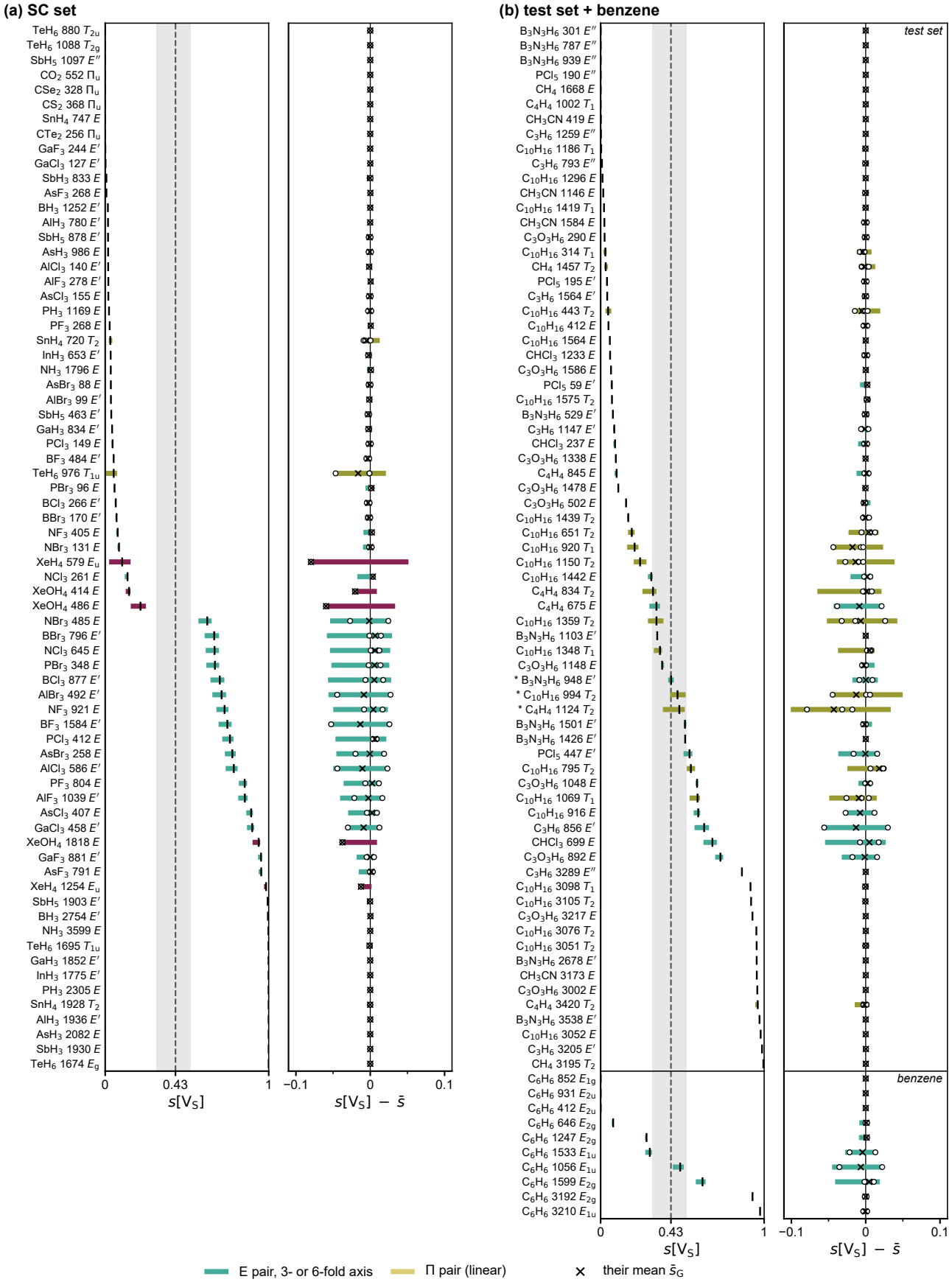

(a) SC set
(b) test set + benzene
test set
benzene
TeH6 880 T2u
TeH6 1088 T2g
SbH5 1097 E″
CO2 552 Πu
CSe2 328 Πu
CS2 368 Πu
SnH4 747 E
CTe2 256 Πu
GaF3 244 E′
GaCl3 127 E′
SbH3 833 E
AsF3 268 E
BH3 1252 E′
AlH3 780 E′
SbH5 878 E′
AsH3 986 E
AlCl3 140 E′
AlF3 278 E′
AsCl3 155 E
PH3 1169 E
PF3 268 E
SnH4 720 T2
InH3 653 E′
NH3 1796 E
AsBr3 88 E
AlBr3 99 E′
SbH5 463 E′
GaH3 834 E′
PCl3 149 E
BF3 484 E′
TeH6 976 T1u
PBr3 96 E
BCl3 266 E′
BBr3 170 E′
NF3 405 E
NBr3 131 E
XeH4 579 Eu
NCl3 261 E
XeOH4 414 E
XeOH4 486 E
NBr3 485 E
BBr3 796 E′
NCl3 645 E
PBr3 348 E
BCl3 877 E′
AlBr3 492 E′
NF3 921 E
BF3 1584 E′
PCl3 412 E
AsBr3 258 E
AlCl3 586 E′
PF3 804 E
AlF3 1039 E′
AsCl3 407 E
GaCl3 458 E′
XeOH4 1818 E
GaF3 881 E′
AsF3 791 E
XeH4 1254 Eu
SbH5 1903 E′
BH3 2754 E′
NH3 3599 E
TeH6 1695 T1u
GaH3 1852 E′
InH3 1775 E′
PH3 2305 E
SnH4 1928 T2
AlH3 1936 E′
AsH3 2082 E
SbH3 1930 E
TeH6 1674 Eg
B3N3H6 301 E″
B3N3H6 787 E″
B3N3H6 939 E″
PCl5 190 E″
CH4 1668 E
C4H4 1002 T1
CH3CN 419 E
C3H6 1259 E″
C10H16 1186 T1
C3H6 793 E″
C10H16 1296 E
CH3CN 1146 E
C10H16 1419 T1
CH3CN 1584 E
C3O3H6 290 E
C10H16 314 T1
CH4 1457 T2
PCl5 195 E′
C3H6 1564 E′
C10H16 443 T2
C10H16 412 E
C10H16 1564 E
CHCl3 1233 E
C3O3H6 1586 E
PCl5 59 E′
C10H16 1575 T2
B3N3H6 529 E′
C3H6 1147 E′
CHCl3 237 E
C3O3H6 1338 E
C4H4 845 E
C3O3H6 1478 E
C3O3H6 502 E
C10H16 1439 T2
C10H16 651 T2
C10H16 920 T1
C10H16 1150 T2
C10H16 1442 E
C4H4 834 T2
C4H4 675 E
C10H16 1359 T2
B3N3H6 1103 E′
C10H16 1348 T1
C3O3H6 1148 E
* B3N3H6 948 E′
* C10H16 994 T2
* C4H4 1124 T2
B3N3H6 1501 E′
B3N3H6 1426 E′
PCl5 447 E′
C10H16 795 T2
C3O3H6 1048 E
C10H16 1069 T1
C10H16 916 E
C3H6 856 E′
CHCl3 699 E
C3O3H6 892 E
C3H6 3289 E″
C10H16 3098 T1
C10H16 3105 T2
C3O3H6 3217 E
C10H16 3076 T2
C10H16 3051 T2
B3N3H6 2678 E′
CH3CN 3173 E
C3O3H6 3002 E
C4H4 3420 T2
B3N3H6 3538 E′
C10H16 3052 E
C3H6 3205 E′
CH4 3195 T2
C6H6 852 E1g
C6H6 931 E2u
C6H6 412 E2u
C6H6 646 E2g
C6H6 1247 E2g
C6H6 1533 E1u
C6H6 1056 E1u
C6H6 1599 E2g
C6H6 3192 E2g
C6H6 3210 E1u
0
0.43
1
s[VS]
−0.1
0
0.1
s[VS] − s̄
E pair, 3- or 6-fold axis
T triple
E pair, 4-fold axis
Π pair (linear)
subspace mean s̄
Gaussian's members
their mean s̄G
ambiguity range (0.31–0.53)
τSB = 0.43